

**Le novità di SQL Server 2025 per
gli sviluppatori**

DATA
SATURDAYS

hn0va
community

Sponsors



About me

- Consulente indipendente
 - SQL Server, PostgreSQL, Fabric, Power BI, AI
- Microsoft Certified Trainer dal 2000
- 6x Microsoft Data Platform MVP

@ ddominici@sqlserverinfo.it

in linkedin.com/in/danilodominici

github.com/ddominici/presentations



DATA
SATURDAYS

The AI-ready enterprise database from ground to cloud

No pricing or licensing changes



SQL Server 2025 editions



Express

Free, entry-level database for small AI, web and mobile apps

Feature highlights

- 50 GB maximum relational databases size **NEW**
- Up to 4 cores of CPU
- Up to 1410 MBs of memory
- Built-in vector search and semantic search **NEW**
- Native JSON type support **NEW**
- External REST endpoint invocation **NEW**
- Change event streaming **NEW**
- Microsoft Entra ID authentication **NEW**
- Mirroring in Fabric **NEW**



Standard

Full featured database with extensive scale for mid-tier applications

Feature highlights

+ Express features

- Up to 32 cores of CPU **NEW**
- Up to 256 GB of memory **NEW**
- Optimized locking **NEW**
- Resource Governor **NEW**
- ZSTD Backup Compression Algorithm **NEW**
- Basic Availability Groups
- Azure SQL Managed Instance link
- Accelerated Database Recovery
- Backups to S3-compatible object storage
- 524 PB maximum relational databases size



Enterprise

Highest performance and scaling for business-critical workloads

Feature highlights

+ Express and Standard features

- Highest core count supported
- Highest memory supported
- Built-in Intelligent Query Processing
- Online and Resumable Index Operations
- Availability Groups: including distributed and contained
- Query Store for readable secondaries
- Persisted Statistics for Secondary Replicas (Query store)

Developer

Free to use for dev/test in non-production environments



Standard Developer **NEW**



Enterprise Developer



Optimized Locking

Eliminate
lock escalation

Lock After
Qualification (LAQ)

Requires
ADR, RCSI,
and db option



ABORT_QUERY_EXECUTION query hint

Abort future
query execution

Stop rogue queries
from harming other users

Query Store Hint



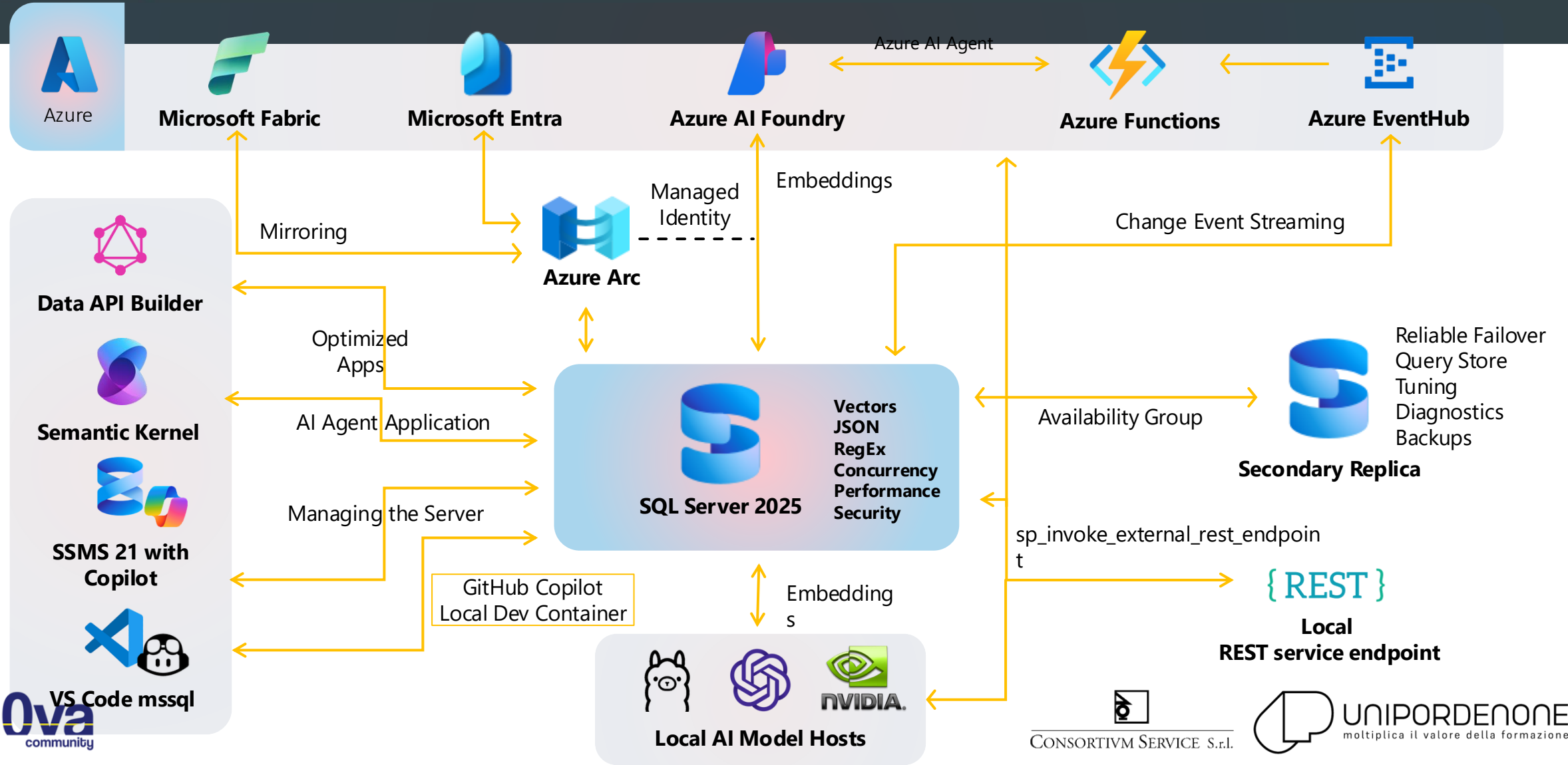
Tempdb Resource Governance

Tempdb space usage can
grow out of control

Need to control and
isolate users

Flexibility to change
policy over time.
Fixed size or %

SQL Server 2025: visione di insieme



Integrazione con l'AI

- ✓ **Ricerche smart** sui dati testuali esistenti
- ✓ Consolidamento di documenti/testi sui quali effettuare **ricerche semantiche**
- ✓ Predisporre **building blocks** – assistenti intelligenti, RAG, Agenti AI
- ✓ Utilizzare l'AI in modo **sicuro e scalabile**
- ✓ **Eliminare la complessità** utilizzando il linguaggio T-SQL

Quali funzionalità introduce SQL Server 2025?



Vector Store

Tipo dati vettoriale nativo e indici
DiskANN*



Ricerca semantica

Vector distance (KNN) e Vector search
(ANN)*



Gestione dei modelli

Definizione nel database dei modelli utilizzati



Gestione degli embeddings

Text Chunking e generazione degli
embeddings (comandi e chiamate REST)



Integrazione con i framework di sviluppo

LangChain, Semantic Kernel, EF Core, Agent
Framework







Accesso spento per default!



Controllo di tutti gli accessi con la sicurezza di SQL Server



Controllo dell'isolamento dei modelli



Scambio dati tra SQL Server e modelli è *cifrato*



RLS, TDE, Dynamic Data Masking, e Auditing



Tabelle *ledger* immutabili per i log delle operazioni degli agenti AI

Scegliere il modello più adatto

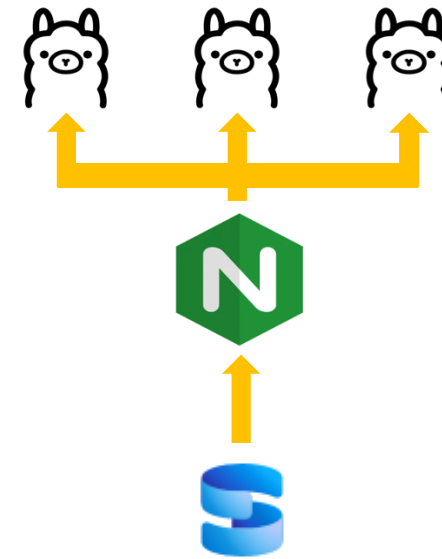
- Hugging Face mantiene una leaderboard dei modelli

Rank (Box...)	Model	Zero-shot	Memory U...	Number of P...	Embedding D...	Max Tokens	Mean (T...	Mean (TaskT...	Bitext ...	Classification	Clustering	Instruction R...
1	llama-embed-nemotron-8b	99%	28629	7B	4096	32768	69.46	61.09	81.72	73.21	54.35	10.82
2	gemini-embedding-001	99%	Unknown	Unknown	3072	2048	68.37	59.59	79.28	71.82	54.59	5.18
3	Qwen3-Embedding-8B	99%	28866	7B	4096	32768	70.58	61.69	80.89	74.00	57.65	10.06
4	Qwen3-Embedding-4B	99%	15341	4B	2560	32768	69.45	60.86	79.36	72.33	57.15	11.56
5	Qwen3-Embedding-0.6B	99%	2272	595M	1024	32768	64.34	56.01	72.23	66.83	52.33	5.09
6	gte-Qwen2-7B-instruct	⚠ NA	29040	7B	3584	32768	62.51	55.93	73.92	61.55	52.77	4.94
7	Linq-Embed-Mistral	99%	13563	7B	4096	32768	61.47	54.14	70.34	62.24	50.60	0.94
8	multilingual-e5-large-instruct	99%	1068	560M	1024	514	63.22	55.08	80.13	64.94	50.75	-0.40
9	embeddinggemma-300m	99%	578	307M	768	2048	61.15	54.31	64.40	60.90	51.17	5.61
10	SFR-Embedding-Mistral	96%	13563	7B	4096	32768	60.90	53.92	70.00	60.02	51.84	0.16
11	text-multilingual-embedding-002	99%	Unknown	Unknown	768	2048	62.16	54.25	70.73	64.64	47.84	4.08

- Usa modelli leggeri
- Esegui inferenze in batch
- Mantieni logging e tracciabilità delle chiamate AI
- Monitora performance e tempi di risposta
- Utilizza cache dei risultati per ridurre le chiamate alle API
 - E quindi costi e latenza
 - Es. Redis

<https://robert-mcdermott.medium.com/large-scale-batch-processing-with-ollama-1e180533fb8a>

- Problema: chiamare un servizio di embedding esterno a SQL Server via REST su HTTPs è limitato dalla banda del backend.
- Con dataset molto grandi i tempi di attesa possono essere molto lunghi..
- Soluzione: bilanciare le chiamate di generazione degli embedding utilizzando istanze multiple di Ollama bilanciate, ad esempio, via Nginx.



Demo



Data API Builder
REST or GraphQL

There's more...



GitHub Copilot for MSSQL



Microsoft Python Driver
for SQL Server



JSON

JSON type, index,
and updated T-SQL
functions

T-SQL

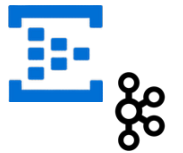
RegEx and other
T-SQL functions

Change Event Streaming*

Consume log change
events

REST API

Connect your data
to **any** REST interface
with HTTPS



{ REST }



gpt-oss



< SQL Server 2025

varchar/nvarchar column

Any index that supports varchar/nvarchar

Various functions for JSON

SQL Server 2025



json data type – binary storage up to 2GB

.modify operator

json index

+ JSON_OBJECTAGG and JSON_ARRAYAGG

+ JSON_CONTAINS

+ JSON_QUERY WITH ARRAY WRAPPER

Save space with compressed storage

Less I/O and page reads



Demo



RegEx
functions

PRODUCT()

BASE64
functions

Fuzzy string match
functions

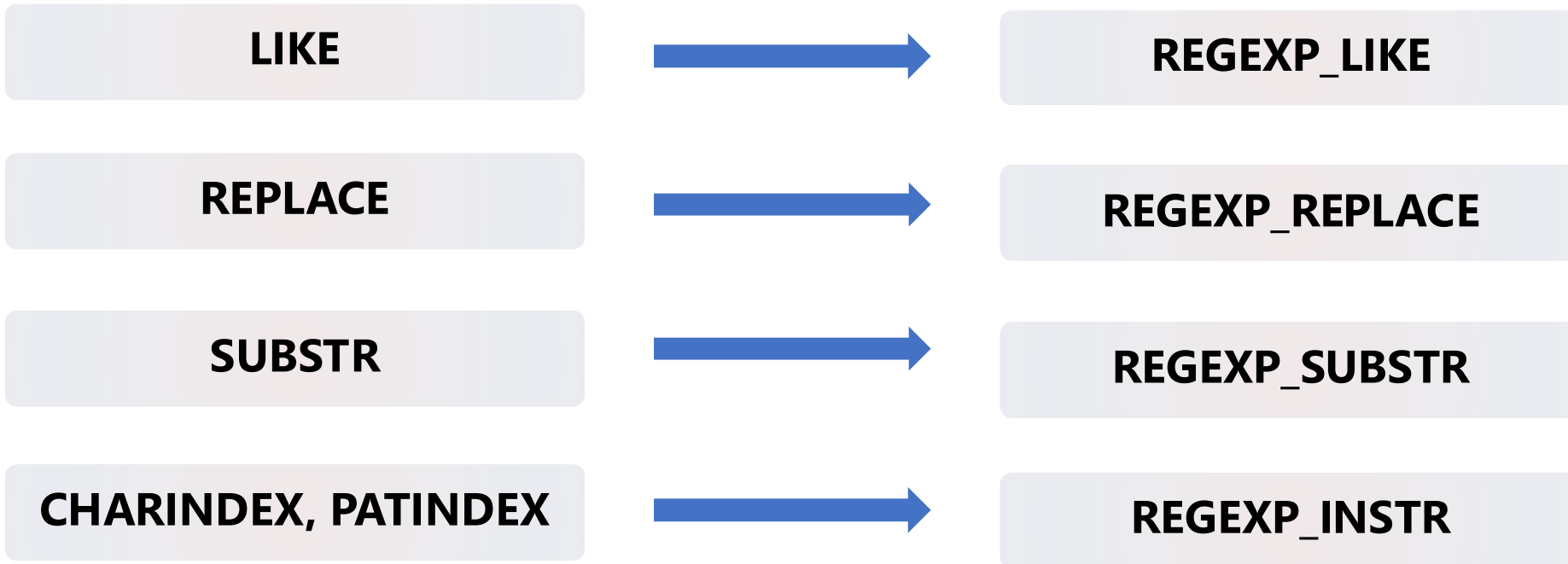
CURRENT_DATE
returns DATE

SUBSTRING()
optional length

UNISTR()

||
string concat
operator

DATEADD() supports
bigint



LEN and REPLACE



REGEXP_COUNT

**CHARINDEX, PATINDEX,
SUBSTR**



REGEXP_MATCHES

**CHARINDEX, REPLACE,
SUBSTR**



REGEXP_SPLIT_TO_TABLE

Demo

Securely avoid roundtrips to send data to a REST endpoint

Off by default

```
EXECUTE @returnValue = sp_invoke_external_rest_endpoint
```

```
[ @url = ] N'url'
```

HTTPS local or remote

```
[ , [ @payload = ] N'request_payload' ]
```

JSON, XML ,TEXT from your data

```
[ , [ @headers = ] N'http_headers_as_json_array' ]
```

```
[ , [ @method = ] 'GET' | 'POST' | 'PUT' | 'PATCH' | 'DELETE' | 'HEAD' ]
```

```
[ , [ @timeout = ] seconds ]
```

```
[ , [ @credential = ] credential ]
```

API key, SAS token, managed identity

```
[ , @response OUTPUT ]
```

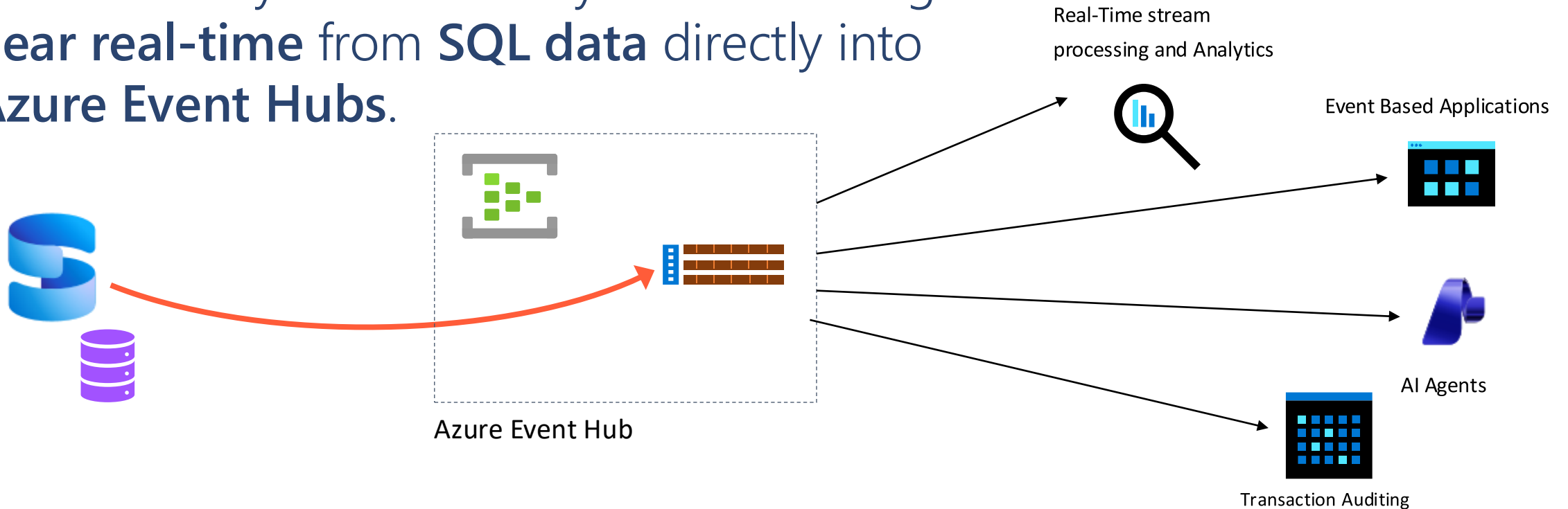
NVARCHAR but often JSON

```
[ , [ @retry_count = ] # of retries if there are errors ]
```

HTTP_EXTERNAL_CONNECTION wait type

Demo

CES enables you to stream your data changes in **near real-time** from **SQL data** directly into **Azure Event Hubs**.



Capturing changes with SQL Server

	Change Tracking (CT)	Change Data Capture (CDC)	Change Event Streaming (CES)
Capture Source	As part of committed transactions (sync)	Committed transactions from transaction log (async). Requires SQL Server Agent	Committed transactions from transaction log (async)
Capture Destination	System table (minimum)	System table (extra logging required)	Event stream ("IOless")
Data Captured	Row identifier (DDL allowed)	Row identifier and data	Row identifier and data (DDL allowed)
Capture Method	Query system tables and user tables. Multiple destinations	Query system tables. Multiple destinations	Process event stream ("IOless"). Single destination
Availability	Immediate (pull-based)	Near real-time (pull-based)	Near real-time (push-based)
SQL effects	Transaction commit	Transaction log truncation	Transaction log truncation
Typical Use Cases	Cache invalidation and sync	Data warehousing and auditing	Event-driven architectures – microservices integration, real-time analytics, cache sync, AI Agents

Cosa fare ora?



Controllare il blog di MS

aka.ms/sqlserver2025blog



Guardare i video (MS)

aka.ms/Build/sql2025mechanics
aka.ms/SQL2025videos



Leggere la documentazione!

aka.ms/sqlserver2025docs



Scaricarlo e provarlo

aka.ms/getsqlserver



Scaricare le demo (MS)

aka.ms/sqlserver2025demos



Scaricare SSMS 22

aka.ms/ssms

That's all folks!